

Pratik Shrivastava

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WORK EXPERIENCE

Dodge AI

Jun. 2025 – Jul. 2025

Machine Learning Intern

Remote, KA

- Built a purchase-intent classifier on 2.75M+ events (22 leakage-safe RFM features), reaching 0.156 PR-AUC $\sim 2\times$ baseline – on a 0.2%-positive class.
- Shipped a hybrid recommender (content + ALS + cold-start) scoring $\sim 24\times$ the MAP@10 of a most-popular baseline on held-out users.
- Used 16-factor ALS embeddings instead of a dense item-similarity matrix (infeasible at 185K products), scaling coverage from 5K to the full catalog.

Siemens Technology & Services Pvt Ltd.

May. 2024 – Aug. 2024

Software Engineering Intern

Remote, KA

- Built RESTful API integration layer for a 3-tier IIoT water-monitoring dashboard with near real-time polling and incremental UI updates.
- Designed GIS map visualizations with colour-coded, per-sensor status markers for quick anomaly spotting.
- Handled missing/inconsistent sensor data gracefully in the parsing layer to keep visualizations from breaking.

PROJECTS

Drape

[Project Link](#)

Python, FastAPI, DINOv2, FAISS, ViT

- Built an image-to-image visual fashion search engine over a 20K-item catalog using YOLOv8 detection + DINOv2 embeddings (768-d) + FAISS search, achieving sub-10ms retrieval latency.
- Designed a neighbor-consensus system generating product descriptions and retailer links directly from retrieval results cutting substantial API calls per search to a handful of parallel fetches.
- Identified a data-integrity flaw in a public benchmark dataset via cross-tabulation analysis, excluding non-unique identifiers from the 20K-item catalog before they could corrupt product metadata.

Harmoniq

[Project Link](#)

Python, FastAPI, Scikit-learn, KNN, Numpy, Pandas

- Built a content-based music recommender over an 89K track Spotify catalog: cosine KNN on a 9-D audio feature embedding.
- Engineered the data pipeline – deduplicated $\sim 114K$ rows to 89K tracks and dropped an autoencoder after measuring that a standardized-feature space gave better cosine-ranking resolution.
- Designed an explainable ranking layer that scores every recommendation on 6 weighted signals and surfaces the closest audio feature as a plain-English reason, with per-artist diversity caps.

Booktures

[Project Link](#)

Python, FastAPI, Scikit-learn, KNN, Numpy, Pandas

- Built a 5-stage PDF-to-illustrated-book pipeline (multi-extractor text parsing + OCR, LLM character extraction, prompt generation, image rendering) driven by a Postgres-backend job queue with pause/resume and restart recovery.
- Engineered a “visual bible” for cross-page character consistency: union-find alias grouping + single-pass trait extraction inlined into every image prompt.
- Abstracted all AI behind 3 pluggable provider layers (LLM/images/storage, 9 backends), so identical code runs on a local GPU using Diffusion models or free tier deployment.

SKILLS

Programming Languages	Python, C++, JavaScript, SQL
Frameworks & Libraries	Node.js, RESTful APIs, FastAPI, Express.js
Databases	MySQL, PostgreSQL, Supabase BaaS
ML	Transformers, ViTs, VLMs, Pytorch, Pandas, Tensorflow, Keras, Matplotlib, Numpy, Pandas
Developer Tools	Git, Github, Postman, VS Code

EDUCATION

Manipal University

Aug. 2022 – Aug 2026

Bachelor of Technology in Computer Science & Engineering

Jaipur, Rajasthan